

Precision, Recall & F1 - All 70 Query Plans

Computed from the same ground truth and prediction data as the accuracy evaluators in build_simple_benchmark_pdf.py. Precision = of what the system returned, how much was correct. Recall = of what truly existed, how much was found. F1 = their harmonic mean. The last column repeats the old single 'overall accuracy' so the gap is visible.

Metric family: detection (frame) = per-frame is-target-present; retrieval (set) = distinct values returned vs present; event (window) = per-window event presence; single-answer / count = one answer, exact match or closeness.

Blank cells: 'n/a' = metric does not apply (single-answer / count query); 'undef' = mathematically undefined (no positive cases in the scored span, so recall/F1 have no denominator).

Cars - naive pipeline

Car Brand (Generic)	retrieval (set)	50.0%	6.7%	11.8%	-
Car Brand Ford	detection (frame)	17.6%	100.0%	29.9%	100.00%
Car Brand Renault	detection (frame)	69.3%	94.4%	80.0%	100.00%
Car Brand Toyota	detection (frame)	41.6%	98.7%	58.5%	100.00%
Car Color (Generic)	retrieval (set)	87.5%	100.0%	93.3%	87.50%
Car Color Red	detection (frame)	51.4%	100.0%	67.9%	100.00%
Car Color Grey	detection (frame)	64.3%	94.3%	76.4%	100.00%
Car Color White	detection (frame)	66.9%	84.0%	74.5%	100.00%
Car Color Blue	detection (frame)	18.1%	100.0%	30.6%	100.00%
License Plate Text Extraction (Generic)	retrieval (set)	28.2%	63.3%	39.0%	24.26%
Specific Plate: QRF8G17	detection (frame)	92.9%	100.0%	96.3%	-
Color + License Plate (Red)	retrieval (set)	11.1%	50.0%	18.2%	10.00%
Most Popular Brand	single-answer	n/a	n/a	n/a	100.00%
Most Popular Color	single-answer	n/a	n/a	n/a	100.00%
Most Popular Brand and Color	single-answer	n/a	n/a	n/a	100.00%
Most Popular Color (Ford)	single-answer	n/a	n/a	n/a	0.00%
Most Popular Brand (Red Cars)	single-answer	n/a	n/a	n/a	0.00%
Unique License Plates	retrieval (set)	30.2%	21.1%	24.8%	14.18%
Repeating License Plates	retrieval (set)	58.2%	49.2%	53.3%	36.36%

Cars - optimized pipeline

Car Brand Generic (R854)	retrieval (set)	61.9%	86.7%	72.2%	56.52%
Car Brand Ford (R854)	detection (frame)	17.4%	98.3%	29.5%	100.00%
Car Brand Renault (R854)	detection (frame)	64.0%	96.9%	77.1%	100.00%
Car Brand Toyota (R854)	detection (frame)	41.5%	97.2%	58.1%	100.00%
Car Color Generic (R480)	retrieval (set)	87.5%	100.0%	93.3%	87.50%
Car Color Red (CFred+R480)	detection (frame)	65.2%	78.9%	71.4%	100.00%
Car Color Grey (CFgrey+R480)	detection (frame)	60.1%	91.6%	72.5%	100.00%
Car Color White (CFwhite+R480)	detection (frame)	60.8%	75.4%	67.3%	100.00%
Car Color Blue (CFblue+R480)	detection (frame)	31.6%	94.7%	47.4%	100.00%
License Plate Recognition (R1120)	retrieval (set)	22.7%	61.1%	33.1%	19.86%
Specific Plate QRF8G17 (R1120)	detection (frame)	85.9%	100.0%	92.4%	100.00%
Color+Plate Red (CFred+R1120)	retrieval (set)	33.3%	50.0%	40.0%	25.00%
Most Popular Brand (R854)	single-answer	n/a	n/a	n/a	100.00%

Most Popular Color (R480)	single-answer	n/a	n/a	n/a	0.00%
Most Popular Brand+Color (R854)	single-answer	n/a	n/a	n/a	100.00%
Most Popular Color Ford (R854)	single-answer	n/a	n/a	n/a	0.00%
Most Popular Brand Red (CFred+R854)	single-answer	n/a	n/a	n/a	100.00%
Unique Plates (R1120)	retrieval (set)	22.7%	61.1%	33.1%	19.86%
Repeating Plates (R1120)	retrieval (set)	62.5%	7.7%	13.7%	7.35%
Red Plate Lookup (CFred+R1120+S3)	retrieval (set)	33.3%	50.0%	40.0%	-

Volleyball - naive pipeline

Repeated Spikers	event (window)	40.0%	100.0%	57.1%	100.00%
Repeated Setters	event (window)	75.0%	75.0%	75.0%	100.00%
Repeated Blockers	event (window)	40.0%	100.0%	57.1%	100.00%
Repeated Diggers	event (window)	60.0%	100.0%	75.0%	100.00%
Motion Category	single-answer	n/a	n/a	n/a	0.00%
Players Moving	event (window)	60.0%	100.0%	75.0%	100.00%
Players Standing	event (window)	100.0%	100.0%	100.0%	100.00%
Players Jumping	event (window)	0.0%	undef	undef	100.00%
Top Actions	single-answer	n/a	n/a	n/a	0.00%
Spike Count	count	n/a	n/a	n/a	0.00%
Set Count	count	n/a	n/a	n/a	98.17%
Block Count	count	n/a	n/a	n/a	0.00%
Set->Spike Events	event (window)	0.0%	undef	undef	100.00%
Set->Block Events	event (window)	0.0%	undef	undef	100.00%
Dig->Set Events	event (window)	0.0%	undef	undef	100.00%
Jump->Spike Events	event (window)	0.0%	undef	undef	100.00%

Volleyball - optimized pipeline

Repeated Spikers (G+R854)	event (window)	40.0%	100.0%	57.1%	100.00%
Repeated Setters (G+R854)	event (window)	37.5%	75.0%	50.0%	100.00%
Repeated Blockers (G+R854)	event (window)	40.0%	100.0%	57.1%	100.00%
Repeated Diggers (G+R854)	event (window)	60.0%	100.0%	75.0%	100.00%
Motion Category (R480)	single-answer	n/a	n/a	n/a	0.00%
Players Moving (R480)	event (window)	60.0%	100.0%	75.0%	100.00%
Players Standing (R480)	event (window)	100.0%	100.0%	100.0%	100.00%
Players Jumping (R480)	event (window)	0.0%	undef	undef	100.00%
Top Actions (R854)	single-answer	n/a	n/a	n/a	0.00%
Spike Count (R854)	count	n/a	n/a	n/a	0.00%
Set Count (R854)	count	n/a	n/a	n/a	0.00%
Block Count (R854)	count	n/a	n/a	n/a	0.00%
Set->Spike Events (R854)	event (window)	0.0%	undef	undef	100.00%
Set->Block Events (R854)	event (window)	0.0%	undef	undef	100.00%
Dig->Set Events (R854)	event (window)	0.0%	undef	undef	100.00%
Jump->Spike Events (R854)	event (window)	0.0%	undef	undef	100.00%